TECHNIK NEST PVT LTD

Summer 2026 Training + Internship Program

Machine Learning & Artificial Intelligence

Internship Task 1

Data Cleaning Using Kaggle Dataset in Google Colab

Submitted By

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Program: Machine Learning & Artificial Intelligence

Organization: Technik Nest Pvt Ltd

Training Cohort: Summer 2026 Training + Internship

Instructor: Abdullah Sajid

Task Report 01

Title: Data Cleaning and Initial Data Exploration Using a Kaggle Dataset in Google Colab

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Objective

The objective of this lab is to learn the basic preprocessing steps required before building any Machine Learning model. This includes downloading a dataset from Kaggle, importing it into Google Colab, examining its structure, identifying missing values and duplicate records, and preparing a clean dataset for future Machine Learning tasks.

Tools Used

- Google Colab
- Python
- Pandas
- NumPy
- Kaggle Dataset

Dataset Information

Dataset Name: Student Math Performance Dataset

Source: Kaggle

Dataset File: student_math_clean.csv

Introduction

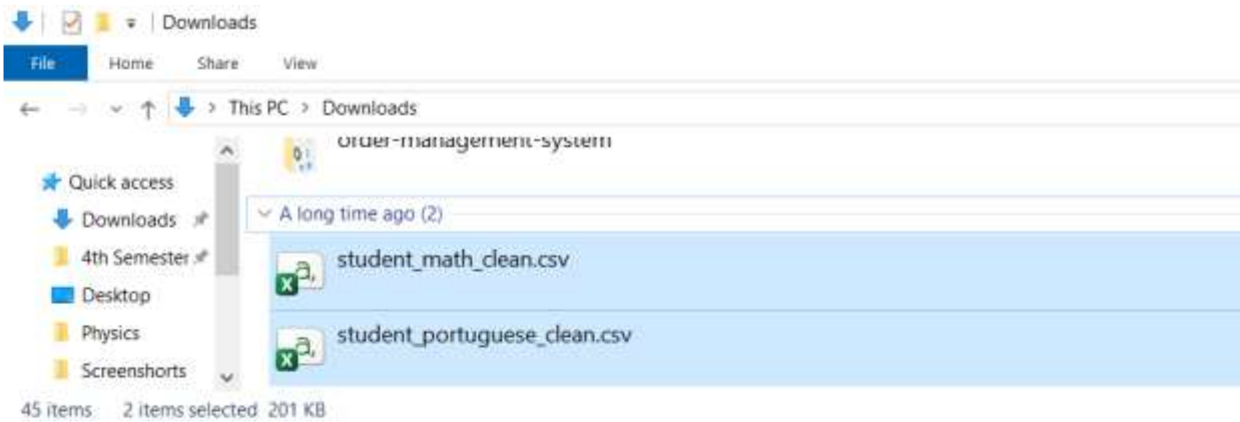
Data cleaning is one of the most important stages in Machine Learning. Before training any model, the dataset must be checked for missing values, duplicate records, incorrect data types, and other inconsistencies. Clean data helps improve the accuracy and reliability of Machine Learning models.

In this lab, a student performance dataset was downloaded from Kaggle and analyzed using Google Colab. Various preprocessing techniques were applied to ensure the dataset was ready for further Machine Learning tasks.

Procedure

Step 1: Download Dataset

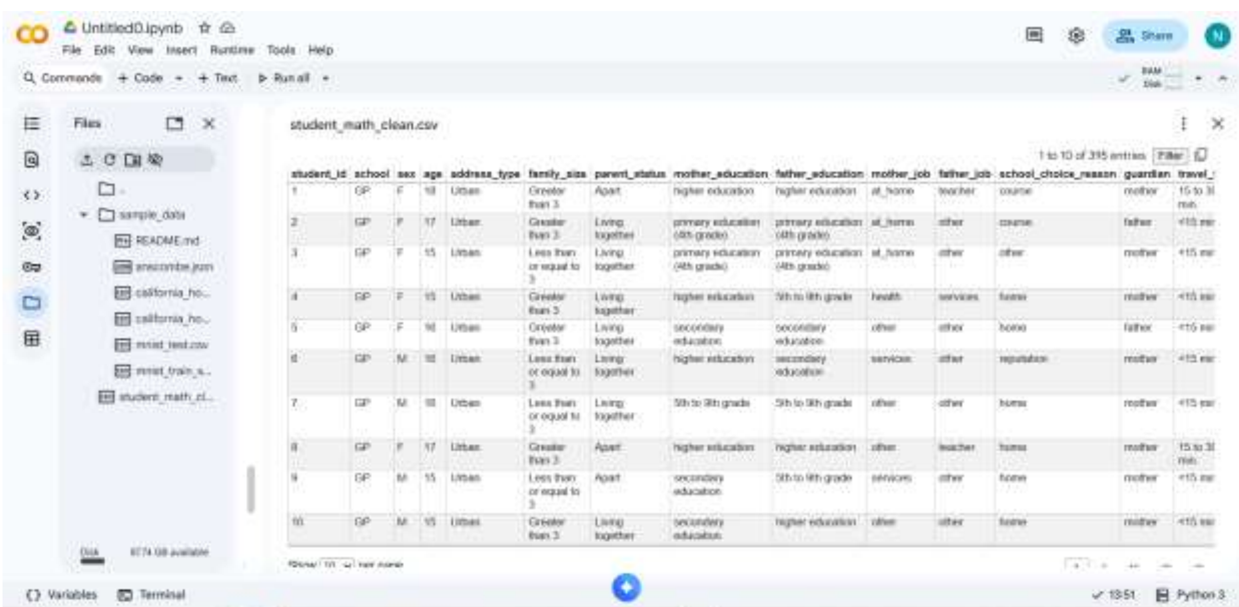
The dataset was downloaded from Kaggle in CSV format.



Step 2: Open Google Colab and Upload Dataset

A new notebook was created in Google Colab.

The downloaded CSV file was uploaded into Google Colab.



Step 3: Import Required Libraries and Load Dataset

- The required Python libraries were imported.
- import pandas as pd
- Import numpy as np
- The dataset was loaded into a Pandas DataFrame.
- df = pd.read_csv("student_math_clean.csv")
- The first five rows of the dataset were displayed.
- df.head()

Observation

The dataset loaded successfully and displayed the first five records.

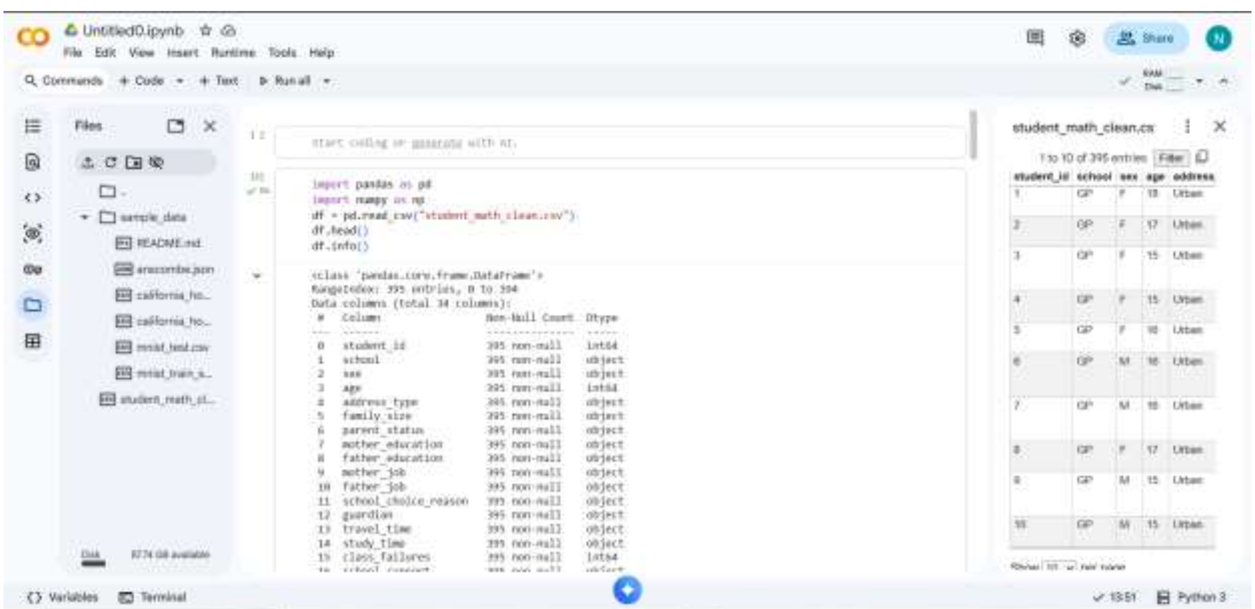
Check Dataset Information

The structure of the dataset was examined.

df.info()

Observation

- Total Records: **395**
- Total Columns: **34**
- Dataset loaded successfully.
- Data types include integers and object (categorical) values.



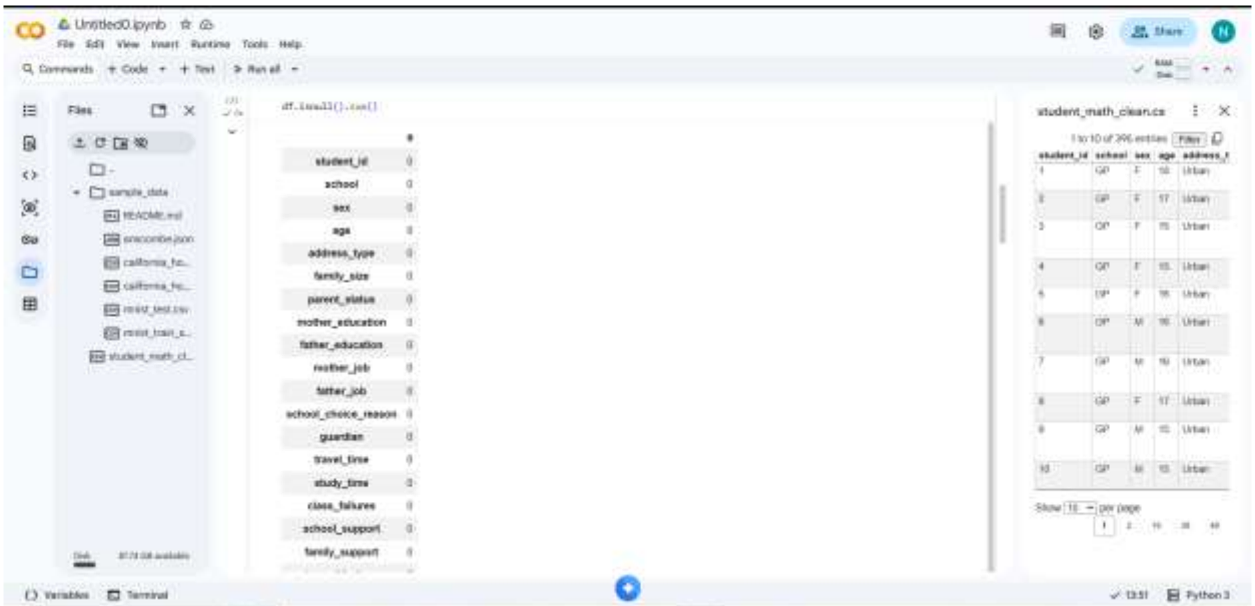
Step 4: Check Missing Values

The dataset was checked for missing values.

df.isnull().sum()

Observation

No missing values were found in any column.



Step5: Check Duplicate Records

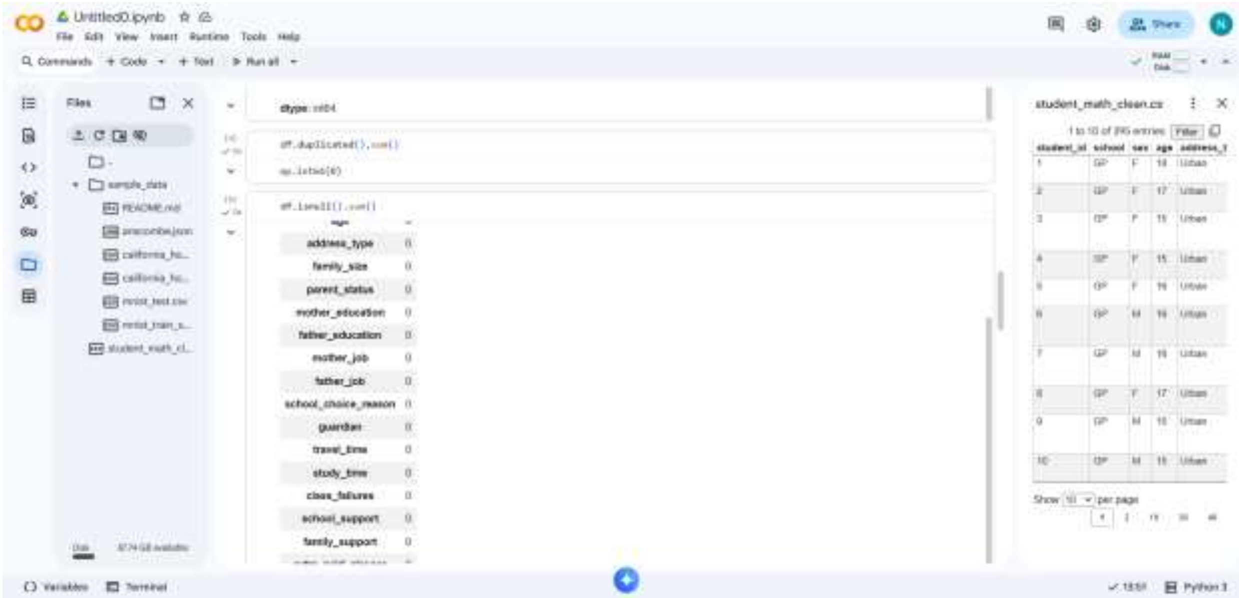
The dataset was checked for duplicate rows.

df.duplicated().sum()

Observation

Output: 0

This indicates that there are no duplicate records.



Step 6: Check Dataset Dimensions

The dataset dimensions were examined.

df.shape

Observation

(395, 34)

Meaning:

- 395 Rows
- 34 Columns

Step 7: Generate Summary Statistics

The statistical summary of numerical columns was generated.

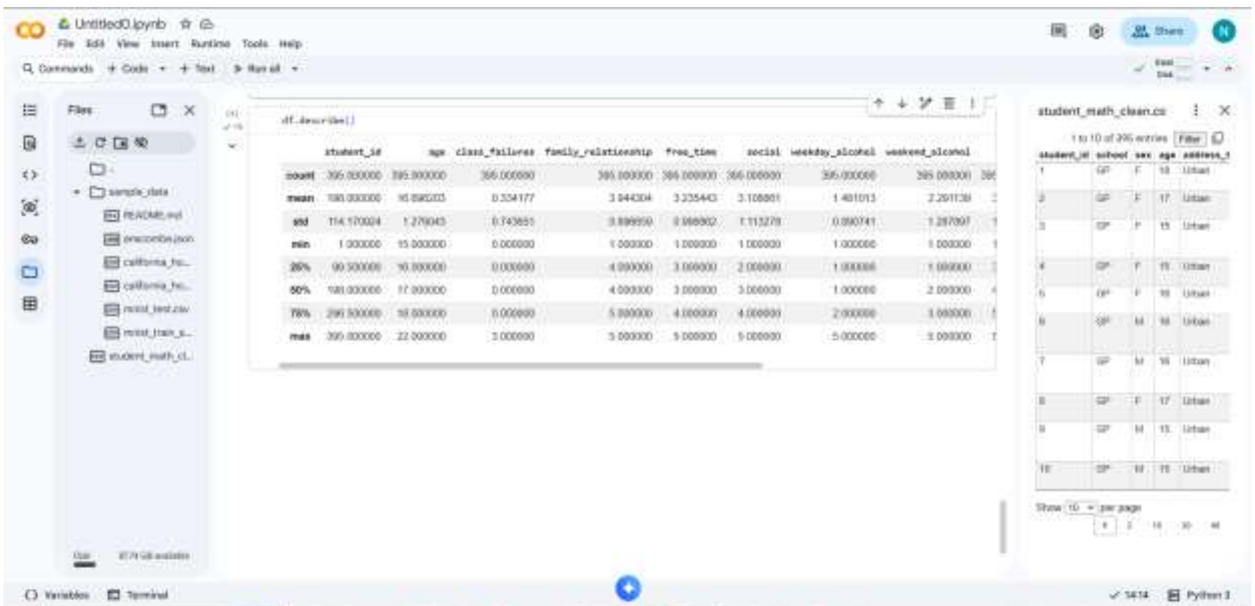
df.describe()

Observation

The summary includes:

- Count
- Mean
- Standard Deviation
- Minimum
- Maximum
- Quartiles

These statistics help understand the distribution of numerical data.



Findings

- The dataset contains **395 records** and **34 attributes**.
- No missing values were detected.
- No duplicate records were found.
- The dataset is already clean and suitable for Machine Learning applications.
- Both numerical and categorical features are present in the dataset.

Conclusion

The Kaggle student performance dataset was successfully imported into Google Colab and thoroughly examined. The dataset was inspected for missing values, duplicate records, data structure, and statistical information. Since no missing or duplicate values were found, the dataset required no additional cleaning and is ready for further Machine Learning tasks such as Linear Regression, Polynomial Regression, and predictive modelling.

Learning Outcomes

After completing this task, I learned how to:

- Download datasets from Kaggle.
- Upload datasets into Google Colab.
- Import Python libraries.
- Load datasets using Pandas.
- Display dataset contents.
- Analyze dataset structure.
- Identify missing values.
- Detect duplicate records.
- Generate descriptive statistics.
- Prepare a dataset for Machine Learning.

References

- Kaggle Dataset
- Google Colab
- Python Documentation
- Pandas Documentation
- NumPy Documentation

Submitted To

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